UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

a) How classification of parallel processor is constructed?

b) Give ONE (1) example for each parallel processor architecture.

QUESTION A2 **(5 marks)**

The Difference Engine was a mechanical computer designed by Charles Babbage in the 19th century. It was one of the earliest examples of a mechanical calculator, built to automatically compute mathematical tables for naval navigation.

a) What device was the successor of the Difference Engine?

b) Describe the difference in function between the Difference Engine and Mechanical Calculating Machine.

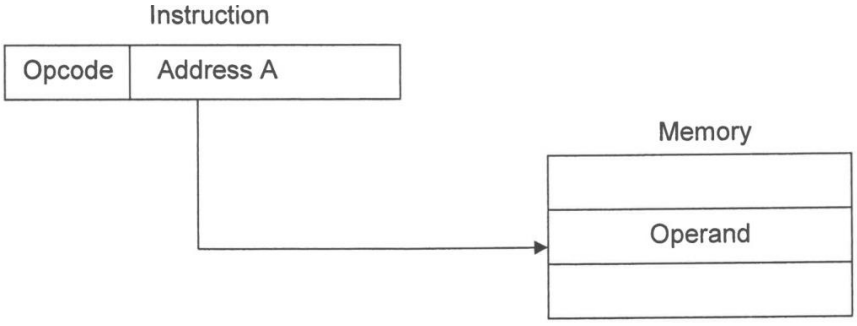
c) Briefly explain the available technology that existed in the computer generation of Difference Engine.

QUESTION A3 **(5 marks)**

Convert 144010 to binary. Show your complete workflow.

QUESTION A4 **(5 marks)**

Given the following Figure 2.



a) Label the addressing technique illustrated in Figure 2.

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b) Explain how the addressing technique in 9(a) is used.

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

QUESTION A5 **(10 marks)**

a) Differentiate between organization and architecture.

b) Explain the function of THREE (3) Central Processing Unit (CPU) components.

QUESTION A6 **(10 marks)**

a) Compare system interconnection and CPU interconnection in terms of:

i. function

ii. physical appearance.

b) Describe the function of the THREE (3) different bus types in a computer system.

QUESTION A7 **(10 marks)**

a) Identify the correct access method based on the following scenario Scenario Access method Readi files. I Accessi c file from a folder. a Streami files line line. AS ecific file block usi an index in a la e file. Accessin toas Jum cific frame or a timestam in the video. tn

b) Rank the following memory types based on the most to the least storage capacity' Main memory, Tape, Ll Cache, Hard drive, ReBister, L3 Cache

c) Discuss the term volatile from the perspective of computer memory.

QUESTION A8 **(10 marks)**

Keyboard is the most common means for computer interaction.

a) Explain TWO (2) types of characters that keyboard provides.

b) Describe THREE (3) categories of input and output module. Give an appropriate example for each category.

QUESTION A9 **(10 marks)**

a) Differentiate between interrupt-driven 1/0 and programmed 1/0.

b) Discuss ONE (1) benefit of programmed 1/0.

c) Discuss TWO (2) disadvantages of multiple interrupts.

QUESTION A10 **(10 marks)**

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: 30 - (-25)

a) Calculate 30 - (-25) using sign and magnitude.

b) Calculate 30 - (-25) using 1's and 2's complement.

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three- address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		